| Subject Code | Subject Name (Lab Oriented Theory Course) | Category | L | T | P | C |
|---|---|---|---|---|---|---|
| IT23731 | Cloud and Big Data Architecture<br>(Common to IT, AIDS, AIML) | PC | 3 | 0 | 2 | 4 |

| |
|---|
| • To understand and appreciate the evolution of cloud from the existing technologies. |
| • To be familiar with cloud computing and have knowledge on the various service models. |
| • To introduce cloud platforms used in industry. |
| • To introduce the concepts of Big Data and Hadoop, and implement map reduce. |
| • To describe the data stream analytics methodologies. |
| • To understand and appreciate the evolution of cloud from the existing technologies. |

| UNIT-I | CLOUD ENABLING TECHNOLOGIES | 9 |
|---|---|---|
| Technologies for Network-Based Systems - System Models for Distributed and Cloud Computing - Implementation Levels of Virtualization - Virtualization Structures/Tools and Mechanisms - Virtualization of CPU, Memory, and I/O Devices - Virtual Clusters and Resource Management - Virtualization for Data-Centre Automation. | | |
| **UNIT-II** | **CLOUD ARCHITECTURE AND SERVICES** | 9 |
| Layered Cloud Architecture Design - NIST Cloud Computing Reference Architecture - Public, Private and Hybrid Clouds - laaS – PaaS – SaaS - Architectural Design of Compute and Storage Clouds -Public Cloud Platforms: GAE, AWS, and Azure. | | |
| **UNIT-III** | **CLOUD PLATFORMS IN INDUSTRY** | |
| Amazon Web Services- Compute Services, Storage Services, Communication Services and Additional Services. Google AppEngine-Architecture and Core Concepts, Application Life-Cycle, cost model. Microsoft Azure- Azure Core Concepts, SQL Azure. | | |
| **UNIT-IV** | **INTRODUCTION TO BIG DATA AND HADOOP** | 9 |
| Introduction to Big Data, Types of Digital Data, Challenges of conventional systems - Web data, Evolution of analytic processes and tools, Analysis Vs reporting - Big Data Analytics, Introduction to Hadoop - Distributed Computing Challenges - History of Hadoop, Hadoop Eco System - Use case of Hadoop – Hadoop Distributors – HDFS – Processing Data with Hadoop – Map Reduce. | | |
| **UNIT-V** | **MINING DATA STREAMS** | 9 |
| Introduction to Streams Concepts – Stream data model and architecture - Stream Computing, Sampling data in a stream – Filtering streams – Counting distinct elements in a stream – Estimating moments – Counting oneness in a window – Decaying window – Real time Analytics Platform (RTAP) applications - case studies - real time sentiment analysis, stock market predictions. | | |
| **Contact Hours:45** | | |

| Description of the Experiments |
|---|
| 1. Install Virtual box /VMware Workstation with different flavors of Linux or windows OS on top of windows7 or 8 |

| | |
|---|---|
| 2. | Install a C compiler in the virtual machine created using virtual box and execute Simple Programs. |
| 3. | Install Google App Engine. Create hello world app and other simple web applications using python/java. |
| 4. | Use GAE launcher to launch the web applications. |
| 5. | Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim. |
| 6. | Find a procedure to transfer the files from one virtual machine to another virtual machine |
| 7. | Find a procedure to launch virtual machine using try stack (Open stack Demo Version) |
| 8. | Install Hadoop single node cluster and run simple applications like word count. |
| 9. | Deploy a Web Application with AWS Elastic Beanstalk. |
| 10. | Launching and configuring EC2 instances. |
| | **Contact Hours : 30** |
| | **Total Contact Hours :75** |

| **Course Outcomes: Students will be able to** |
|---|
| ● Learn the key and enabling technologies that help in the development of the cloud. |
| ● Develop the ability to understand and use the architecture of compute and storage cloud, service and delivery models. |
| ● To understand cloud platforms usage in industry. |
| ● Understand the usage scenarios of Big Data Analysis and Hadoop framework and apply Mapreduce over HDFS. |
| ● Apply stream data models for mining big data |

**SUGGESTED ACTIVITIES**

- Problem Based Learning
- Flipped classroom
- Circuit Design using Simulator
- Conceptual Online Quiz

**SUGGESTED EVALUATION METHODS**

- Continuous Assessment Test
- Online Quiz Assignments
- Offline Assignments
- Experiment based VIVA

**Text Book(s):**

1. Kai Hwang, Geoffrey C. Fox, Jack G. Dongarra, "Distributed and Cloud Computing, From Parallel

| | |
|---|---|
| | Processing to the Internet of Things", Morgan Kaufmann Publishers, First Edition, 2013. |
| 2. | Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley India; Second Edition , 2019. |
| 3. | Tamara Munzner, "Visualization Analysis and Design", AK Peters Visualization Series, CRC Press, Nov. 2014. |
| 4. | Anand Rajaraman and Jeffrey David Ulman, "Mining of Massive Datasets", Cambridge University Press, Second Edition , 2016 |
| 5. | Jiawei Han, Micheline Kamber —Data Mining Concepts and Techniques, Fourth Edition, Elseiver, November 2022. |

| **Reference Books(s) / Web links:** |
|---|
| 1. Gerardus Blokdyk, AWS Certified Solutions Architect A Complete Guide - 2020 Edition. |
| 2. Iman Ghanizada, "Google Cloud Certified Professional Cloud Arcitect", McGraw-Hill Education Publisher, 23rd April 2021. |
| 3. Dr. Jugnesh Kumar , Dr. Anubhav Kumar , Dr. Rinku Kumar, "Big Data and Analytics: The key concepts and practical applications of big data analytics", 1st Edition, BPB Publications, March 2024 |

**CO-PO-PSO Mapping**

| **PO/PSO** / **CO** | **PO1** | **PO2** | **PO3** | **PO4** | **PO5** | **PO6** | **PO7** | **PO8** | **PO9** | **PO10** | **PO11** | **PO12** | **PSO1** | **PSO2** | **PSO3** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| IT23731.1 | 3 | 3 | - | 2 | 3 | 2 | - | - | 1 | - | 1 | 1 | 3 | 2 | 2 |
| IT23731.2 | 3 | 3 | - | 2 | 3 | 2 | - | - | 1 | - | 1 | 1 | 3 | 2 | 2 |
| IT23731.3 | 3 | 3 | - | 2 | 3 | 2 | - | - | 1 | - | 1 | 1 | 3 | 2 | 2 |
| IT23731.4 | 3 | 3 | - | 2 | 3 | 2 | - | - | 1 | - | 1 | 1 | 3 | 2 | 2 |
| IT23731. 5 | 3 | 3 | - | 2 | 3 | 2 | - | - | 1 | - | 1 | 1 | 3 | 2 | 2 |
| Average | 3 | 3 | - | 2 | 3 | 2 | - | - | 1 | - | 1 | 1 | 3 | 2 | 2 |

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

No correlation: “-“